

Agentic AI Training Recommendation Platform

1. Problem Statement

Centralized automated platform to collect software(CSE and related)/hardware (ECE and related) courses from Indian institutes, analyze curricula using AI, recommend training, route recommendations for administrator approval, notify colleges, follow up, and track outcomes.

2. Targeted Users

Students; Colleges (HODs, Placement Coordinators); Universities; Training Providers; Corporate L&D; HR Teams; Administrators.

3. Project Features

- Automated college discovery and web crawling
- Curriculum extraction from HTML/PDF
- AI skill extraction and technology mapping
- Personalized training recommendations
- Administrator review/approval workflow
- Automated email/notification to HODs and placement coordinators
- Follow-up reminders and meeting scheduling
- Dashboards and analytics
- Continuous learning from feedback

4. List of Agents

1. College Discovery Agent
2. Web Crawling Agent
3. Document Parsing Agent
4. Course Intelligence Agent
5. Skill Mapping Agent
6. Recommendation Agent
7. Validation Agent
8. Approval Agent
9. Communication Agent
10. Follow-up Agent
11. CRM Agent
12. Reporting & Analytics Agent

5. End-to-End Workflow

Discover Colleges → Crawl Websites → Parse Curriculum → Extract Skills → Generate Training Recommendation → Administrator Review → Approve/Modify → Notify Colleges → Follow-ups → Training Delivery → Feedback → Analytics & Continuous Learning

6. Architecture

6a. H

High Level Architecture

User Portal → API Gateway → Agent Orchestrator → AI/Recommendation Engine → PostgreSQL + Vector DB + Object Storage → Dashboards

6b. Agent Architecture

Orchestrator coordinates crawler, parser, intelligence, skill mapping, recommendation, approval, communication, follow-up, reporting agents.

6c. Database Architecture

PostgreSQL: colleges, courses, subjects, users, training, approvals, communications, feedback.

Vector DB: embeddings.

Object storage: PDFs, HTML, reports.

6d. API Architecture

REST APIs for colleges, courses, recommendations, approvals, communications, dashboards; LLM/Embedding APIs; authentication.

6e. Folder Structure

```
project/  
  agents/  
  api/  
  workflows/  
  services/  
  database/  
  crawlers/  
  parsers/  
  rag/  
  frontend/  
  tests/  
  docs/  
  logs/  
  data/
```